## the analogy that built chemical engineering

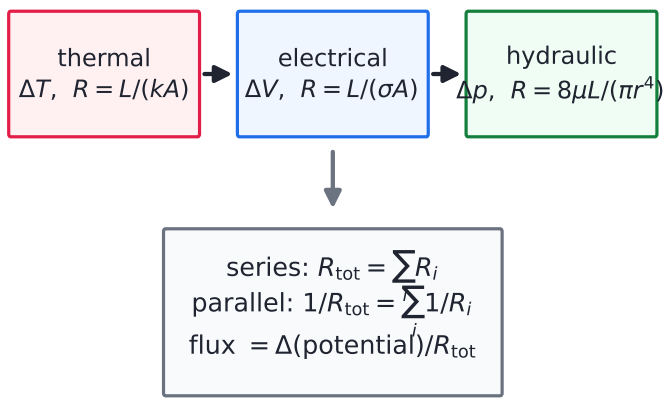
**Reynolds · Prandtl · Chilton–Colburn · Bird–Stewart–Lightfoot**

heat — $q = -k\nabla T$ — potential: $T$ — flux ratio: Nu — Fourier

mass — $J = -D\nabla c$ — potential: $c$ — flux ratio: Sh — Fick

momentum — $\tau = \mu\nabla u$ — potential: $u$ — flux ratio: $C_f$ — Newton

$$\text{flux} = -(\text{transport property}) \times \nabla(\text{potential})$$

## circuit thinking

**every *linear* steady transport problem is a resistor network — which is why finite-volume codes and bond graphs look the same**